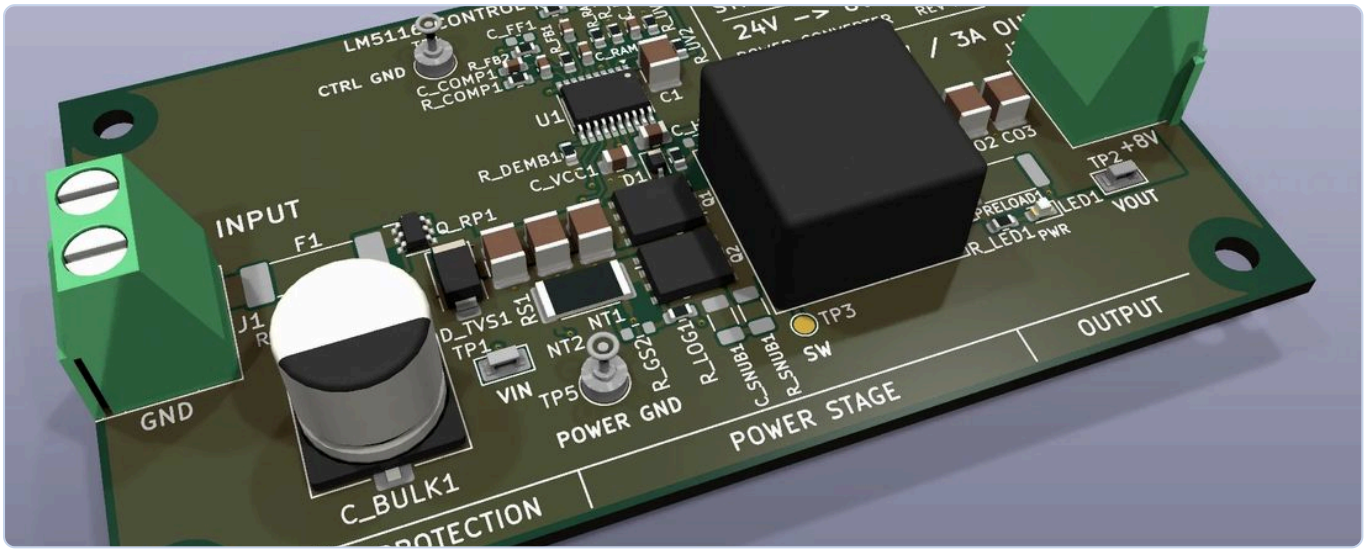
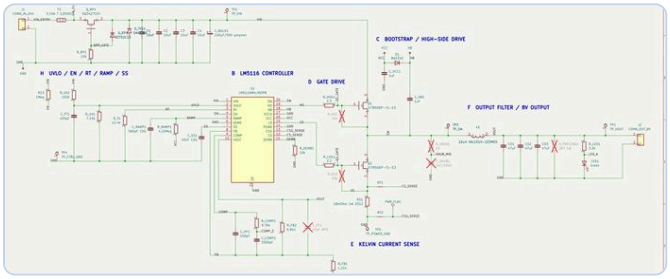
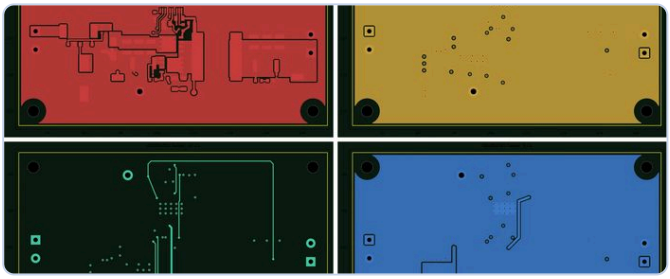
LM5116 – Synchronous Buck Converter 24 V to 8 V / 3 A (KiCad, Prototype Rev A)

Hardware frozen, not manufactured, not measured TI LM5116 · KiCad 10 · LTspice







A four-layer, non-isolated synchronous buck converter built around the TI LM5116, with a 24 V input and an 8 V, 3 A output. Fully designed in KiCad 10, with frozen manufacturing data, audit reports, LTspice simulations and a bring-up plan. The prototype has not been built yet, and all measurements are still to come.

KEY POINTS

- Synchronous buck converter 24 V to 8 V / 3 A at about 251 kHz
- Multi-stage input protection: fuse, reverse-polarity protection, TVS diode
- ERC and DRC clean, LTspice simulation, test plan T00 to T16

KEY DATA

TECHNOLOGY
TI LM5116 · KiCad 10 · LTspice

PLATFORM
Circuit board, 4 layers

STATUS
Hardware frozen, not manufactured, not measured

ACCESS
Files and source code on request

More in the technical portfolio
nguyennhando.github.io/my-electronics-blog/?post=lm5116-buck-converter-24v-8v
Source code and design files are available on request: donguyennhan.de@gmail.com